BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

MATHS PAPER-10th — ANNUAL EXAM-2021 — SECTION-A

Time Allowed: 15 mints | **Maximum Marks:** 12

Note: Q 1: Choose the correct options.

Q1. Let be a function from A to B, then f is onto fraction if...

(A) Range f=A
(B) Range f = B
(C) Range f ≠ A
(D) Range f ≠ B

Q2. A commutative frequency curve is called.....

(A) histogram
(B) ogive
(C) pie chart
(D) frequency polygon

Q3. Mode of the data 5, 5 7, 9, 9, 9, 9, 11, 11, 11, 11, 12, and 12 is.....

(A) 9
(B) 11
(C) both 9 and 11
(D) both 5 and 12

Q4. Circumference of a circle having radius is.....

(A) 44/7
(B) 22/7
(C) 44
(D) 22

Q5. $\sqrt{(1 - \sin^2\theta)}$ = ...

(A) cos θ
(B) sin θ
(C) cot θ
(D) tan θ

Q6. The line which meets the circle at one point is.....

(A) chord
(B) secant
(C) tangent
(D) diameter.

Q7. Angle in a same-circle is a/an .....angle.

(A) Acute (B) right

(C) obtuse (D) interior

Q8. AUB = ...

(A) {x/x∈ A U x ∈B} (B) {x/x∈ A ∩ x ∈B}

(C) {x/x∈ A U x ∉B} (D) {x/x∈ A ∩ x ∉ BB}

Q9. The sum of (x + 1) and the reciprocal of (x -1) is.....

(A) x/(x-1) (B) x/(x+1)

(C) $x^2/(x-1)$ (D) 2x/(x-1)

Q10. If in a quadratic equations $b^2 - 4ac < 0$ the roots will be..

(A) equal and imaginary (C) equal and real

(C) unequal and imaginary (D) unequal and real

Q11. If α and β be the roots of the equation $ax^2 + b + c = 0$ then the product roots will be.....

(A) [illegible in source scan] (B) c/a

(C) -b/a (D) -c/a

Q12. If x, y and z is in continued proportion, then .....

(A) x/y = y/z (B) $z/x = x^2/y^2$

(C) $x/z = y^2/r^2$ (D) $z/a = b^2$

Q13. If x ∝ 1/y and y ∝ 1/z the .....

(A) 1/z (B) x ∝ z

(C) xy ∝ z (D) xz ∝ y

Q14. $(x^2+x-3)/(x^3+x^2-x+1)$ is ..... Fraction.

(A) proper rational (B) proper irrational

(C) improper rational (D) improper rational

Q15. If (A∩B). Ø. Then A and B are ..... Sets.

(A) Empty (B) equal

(C) overlapping (D) disjoint.

SECTION-B

Time Allowed: 2.45 hrs. | **Maximum Marks:** 32

Q2. (Marks 32) Q 2: Answer any Nine (9) parts. Each part carries 4 marks.

i. Solve $x^2 + 3x - 10 = 0$ by factorization method.

ii. Solve $\sqrt{(6x-5)} = 3x - 10$

iii. Show that $(1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^8) = 1$

iv. If α, β are the roots of $3x^2 - 2x + 5 = 0$, find the equation whose roots are $\alpha/\beta, \beta/\alpha$.

v. Find two consecutive positive integers whose product is 72.

vi. If $3x - 1, 4, 35$ are continued proportional, find the value of x.

vii. Prove that $a/b = c/d = e/f = (a+c+e)/(b+d+f)$

viii. Find the partial fractions of $(3x+2)/(x^2-x-2)$

ix. If U = {a, b, c, d, e, f, g}, and B = {a, b, c} and C = {c, d, e} then prove that (AUB)' = A'∩ B'

x. If x = {1, 2, 3, 4} and y = {5, 6, 7, 8} then write (i) onto function from y on x (ii) Bijective function from y on x

i. Prove that $(1 - \sin\theta)/\cos\theta$

SECTION-C

Maximum Marks: 21

Note: NOTE: Attempt any three (3) questions. All carries equal marks.

Q3. Show that the two triangles drawn to a circle from a point outside it are equal in length?

Q4. prove that the equal chords of congruent circles subtend equal angles at the corresponding centers?

Q5. Show that the angle in semi-circle is a right angle?

Q6. Construct a triangle ABC such that mAB = 3 cm, mAC = 4 cm and m∠ A =60°. Draw circumcircle of this triangle.